FINITE TYPE AFTER REDUCTION IMPLIES TOPOLOGICALLY OF FINITE TYPE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof builds the candidate polynomial map from reduced generators plus ideal generators, expands elements of powers of b into images plus deeper powers by induction, and concludes density from the definition topology of b

Editorial note. The label lemma-Noetherian-finite-type-red names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the condition of lemma-finite-type-red of the same chapter: B/b is generated over A by finitely many classes, where b is the ideal of topologically nilpotent elements
- b is an ideal of definition of the weakly admissible Noetherian ring B
- the definition of topologically of finite type via density of the image of a polynomial map

Source reference. Original reference 0GC2: sources/stacks-restricted/pinned-original-source.tex, source lines 2631–2631. Prerequisite pointer invoked at proof line(s) 2751; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\varphi : A \to B$ be an arrow of $WAdm^{Noeth}$ satisfying the condition defined in Lemma 0GC2. Then $A \to B$ is topologically of finite type.*

Proof. Let $\mathfrak{b} \subset B$ be the ideal of topologically nilpotent elements. Choose $b_1, \ldots, b_r \in B$ which map to generators of $B/\mathfrak{b}$ over A. Choose generators $b_{r+1}, \ldots, b_s$ of the ideal $\mathfrak{b}$. We claim that the image of

$$\varphi : A[x_1, \ldots, x_s] \longrightarrow B, \quad x_i \longmapsto b_i$$

has dense image. Namely, if $b \in \mathfrak{b}^n$ for some $n \geq 0$, then we can write $b = \sum b_E b_{r+1}^{e_{r+1}} \ldots b_s^{e_s}$ for multiindices $E = (e_{r+1}, \ldots, e_s)$ with $|E| = \sum e_i = n$ and $b_E \in B$. Next, we can write $b_E = f_E(b_1, \ldots, b_r) + b'_E$ with $b'_E \in \mathfrak{b}$ and $f_E \in A[x_1, \ldots, x_r]$. Combined we obtain $b \in \mathrm{Im}(\varphi) + \mathfrak{b}^{n+1}$. By induction we see that $B = \mathrm{Im}(\varphi) + \mathfrak{b}^n$ for all $n \geq 0$ which mplies what we want as $\mathfrak{b}$ is an ideal of definition of B. $\square$

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